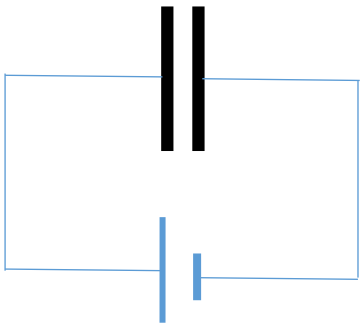
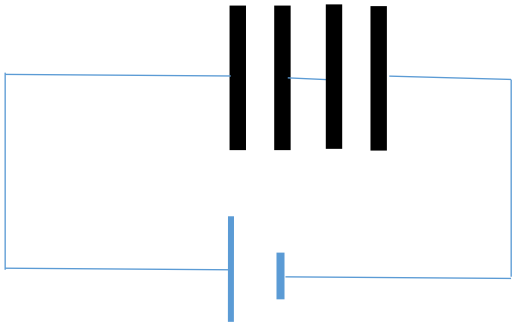
**1- Masala**

1- Hol



Kondensator sig'imi c va manba kuchlanishi U bo'lsa

1- Holda $W_1 = \frac{CU^2}{2}$ ga teng bo'ladi.



2- holda esa $CU_1 = 2CU_2$

$$U_2 = \frac{U_1}{2}$$

$$U_1 + U_2 = U$$

$$U_1 + \frac{U_1}{2} = U$$

$$U_1 = \frac{2U}{3}$$

$$W_1{}' = \frac{CU_1{}^2}{2} = \frac{C\left(\frac{2U}{3}\right)^2}{2} = \frac{4}{9} * w_1$$

$$javob: W_1{}' = \frac{4}{9} * W_1$$

@olimpdep

**2- Masala**

Masala sharti bo'yicha birinchi jism yerga urilish vaqtida tezligi $v_1 = \sqrt{2gh}$

Ikkkinchi jismning yo'lning yarmidagi tezligi esa $v_2' = \sqrt{gh}$

Ikkinchi jism yupqa to'siqqa urilgach tezligi $v_2'' = 0{,}9 * \sqrt{gh}$

Yerga urilish vaqtida esa $v_2 = \sqrt{v_2''^2 + gh} = \sqrt{1{,}81gh}$

$$\frac{v_1}{v_2} = \frac{\sqrt{2gh}}{\sqrt{1{,}81gh}} = \sqrt{\frac{200}{181}}$$

Javob: $\frac{v_1}{v_2} = \sqrt{\frac{200}{181}}$

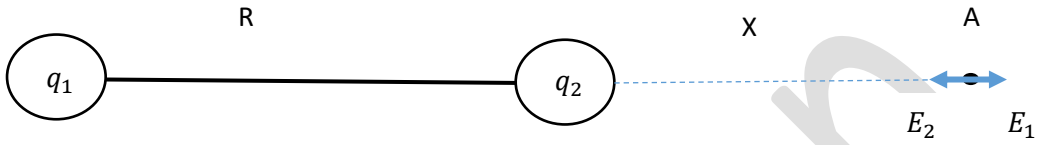
@olimpdep

**3- Masala**

Berilgan

$q_2 = -16\ \text{nC}$ $R = 0{,}4\ m$

$q_1 = 36\ \text{nC}$ R+X=?




$$E_n = E_1 - E_2 = 0$$

$$E_1 = E_2$$

$$E_1 = \frac{k * |q_1|}{(R+x)^2}$$

$$E_2 = \frac{k * |q_2|}{x^2}$$

*bu ikki tenglamadan*

$$\frac{k * |q_1|}{(R+x)^2} = \frac{k * |q_2|}{x^2}$$

$$\sqrt{\frac{|q_1|}{(R+x)^2}} = \sqrt{\frac{|q_2|}{x^2}}$$

$$\frac{\sqrt{|q_1|}}{R+x} = \frac{\sqrt{|q_2|}}{x}$$

Bu tenglamadan X masofani aniqlasak

X=0,8 m= 80 sm

R+x=120 sm

Javob : 120 sm

@olimpdep

Fan olimpiadalari bo‘yicha iqtidorli o‘quvchilar bilan ishlash **DEPARTAMENTI**

**4- Masala**

$q_1' = 4q_1$ $\qquad F_1 = \frac{k\, q_1 q_2}{{R_1}^2}$

$R_2 = \frac{R_1}{2}$ $\qquad F_2 = \frac{k\, q_1' q_2}{\varepsilon\, {R_2}^2}$

$\varepsilon = 16$

$\frac{F_2}{F_1} = ?$ $\qquad \frac{F_2}{F_1} = \frac{k\, q_1 q_2}{{R_1}^2} : \frac{k\, q_1' q_2}{\varepsilon\, {R_2}^2}$

$\frac{F_2}{F_1} = \frac{4*4}{16} = 1$ demak o'zgarmaydi

$javob: o'zgarmaydi$

@olimpdep

**5- Masala**

Berilgan

U=300 V

m=1000 kg

h=15 m

t= 60 s

$\eta = 80\%$

$I$ =?

$$\eta = \frac{mgh}{IUt} * 100\%$$

$$I = \frac{mgh}{\eta * Ut} * 100\%$$

$$I = \frac{100\ kg * 10\left(\frac{N}{kg}\right) * 15\ m}{80\% * 300\ V * 60\ s} 100\% = 3.125\ A$$

Javob: $I = 3.125\ A$

@olimpdep